TECHNICAL REPORT

SMART ENGINE COOLING SYSTEM USING FUZZY LOGIC CONTROL

SUBJECT

Fuzzy Logic and Genetic Programming (BE05000461)

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in Automobile Engineering Department

L. D. COLLEGE OF ENGINEERING, AHMEDABAD GUJARAT TECHNOLOGICAL UNIVERSITY, AHMEDABAD AUGUST, 2026

Design, Rule Formulation, Telemetry Inference, Centroid Defuzzification, and Performance Analysis

Domain	Methodology	Controller Architecture	Status
Automotive Embedded Systems	**Mamdani Fuzzy Inference (FIS)**	**3-Input, 1-Output Engine**	**Complete Technical Specification**

1. Problem Statement

Internal combustion engines and high-performance vehicle powertrains operate under continuously dynamic thermal stress caused by varying engine speeds, varying vehicular loads, and extreme ambient meteorological conditions. Regulating engine block and coolant loop temperatures is vital for thermal efficiency, tailpipe emissions compliance, fuel economy, and mechanical component longevity.

1.1 Limitations of Conventional Thermostatic & On/Off Control

Traditional automotive thermal management systems depend heavily on mechanical thermostats and binary hysteresis controllers (e.g., *"IF Coolant Temperature > 90°C, THEN Radiator Fan = ON for 10 minutes"*). In modern operating contexts, this binary approach suffers from acute engineering shortcomings:

- **Thermal Shock & Micro-Fracturing:** Sudden switching of high-capacity cooling fans injects sudden, steep thermal gradients into the aluminum cylinder head and block, leading to material expansion fatigue and micro-cracking over extended duty cycles.
- **Electromechanical Actuator Wear:** A sharp crisp threshold causes hunting behavior and cycling around boundary points (e.g., oscillating across 89.9°C and 90.1°C), burning relay contacts and deteriorating DC fan motor brushes.
- **Context Ignorance (Environmental Decoupling):** Crisp single-input thermostats fail to recognize ambient conditions. Running a radiator fan for a rigid 10 minutes when driving at high speed in freezing ambient air (10°C) results in severe over-cooling, poor oil viscosity, and excess emissions. Conversely, running the fan for that same duration during stop-and-go traffic in desert heat (40°C) produces engine thermal saturation and boil-over.
- **Parasitic Electrical Load:** Rigid ON/OFF operation draws significant alternator power regardless of whether a minor cooling assist would have safely restored equilibrium, compromising battery state-of-health and fuel economy.

1.2 The Fuzzy Logic Solution Paradigm

Fuzzy Logic resolves these vulnerabilities by transitioning from classical crisp binary sets (where membership is strictly $\mu \in \{0, 1\}$) to continuous fuzzy sets ($\mu \in [0, 1]$). Human automotive calibration engineers naturally make expert qualitative decisions: *"If the engine block is moderately hot, the coolant temperature is creeping high, and the outside air is scorching, run the fan for a prolonged period."* Fuzzy logic embeds this nuanced, multi-variable heuristic expertise into an embedded digital controller without requiring complex, non-linear thermodynamic differential equations.

> **■ CRISP VS. FUZZY THERMAL DYNAMICS**
> A crisp controller treats 89.9°C as 'NOT HOT' (μ = 0) and 90.1°C as 'FULLY HOT' (μ = 1). A 0.2°C telemetry fluctuation triggers full actuator cycling. In contrast, fuzzy logic models smooth overlap: at 85.0°C, μ_Hot = 0.35, and at 92.0°C, μ_Hot = 0.78, eliminating chatter and maintaining thermal equilibrium.

2. Objectives

The primary objective of this project is to architect, formulate, implement, and validate an intelligent, multi-input single-output (MISO) Mamdani Fuzzy Inference System (FIS) for automotive radiator fan management. The specific functional and design objectives comprise:

- **Multi-Sensor Thermal Telemetry Fusion:** Integrate three distinct sensor telemetry streams—Engine Temperature, Coolant Temperature, and Ambient/Environmental Temperature—into a unified decision matrix.
- **Rigorous Mathematical Membership Function Modeling:** Design customized triangular and trapezoidal membership functions across all linguistic terms with calibrated overlap zones to guarantee smooth output interpolation.
- **Expert Heuristic Rule Base Formulation:** Develop an automotive expert IF-THEN rule base employing Zadeh min-intersection operators that realistically captures cross-variable thermal interactions.
- **End-to-End Mamdani Inference Pipeline Implementation:** Formulate the complete 4-stage pipeline: Fuzzification, Mamdani Min Antecedent Evaluation & Implication Clipping, Max Aggregation, and Centroid Defuzzification.
- **Empirical Numerical Verification:** Execute a complete manual and algorithmic trace of a severe real-world thermal telemetry snapshot (Engine Temp = 92°C, Coolant Temp = 88°C, Ambient Temp = 35°C), verifying exact centroid convergence against weighted approximation benchmarks.
- **Actuator Optimization & Energy Longevity:** Deliver smoothly modulated fan run duration (0 to 30 minutes) to eliminate thermal shock, mitigate parasitic alternator draw, and extend electromechanical actuator lifespan.

3. System Design

The smart cooling controller is engineered around a four-stage Mamdani Fuzzy Inference System (FIS) embedded in the vehicle's Powertrain Control Module (PCM/ECU). The architectural topology connects real-time thermistors and ambient sensors through analog signal conditioning to the fuzzy inference engine, which commands the pulse-width modulated (PWM) radiator cooling fan.

3.1 Fuzzy Inference System (FIS) Architecture

The internal FIS pipeline follows a deterministic four-block execution sequence:

- **Stage 1: Fuzzification Interface:** Accepts real-time crisp telemetry signals (°C) from analog thermistors, evaluates boundary conditions, and computes membership degrees (μ) across defined linguistic sets.
- **Stage 2: Knowledge Base & Rule Inference:** Contains the domain rule base and evaluates active rule antecedents using Zadeh min-operator t-norms. Generates rule firing strengths (w_i) and applies Mamdani implication (min-clipping) to output fuzzy sets.
- **Stage 3: Fuzzy Output Aggregation:** Merges the individual clipped output membership areas from all active rules using the Zadeh max-operator s-norm to construct a combined output geometric envelope.
- **Stage 4: Defuzzification Interface:** Calculates the continuous mathematical centroid (Center of Gravity) of the aggregated fuzzy envelope, yielding an exact crisp control command: Radiator Fan Duration in minutes.

3.2 Definition of System Variables & Universes of Discourse

The controller maps three continuous input variables to one continuous actuator output variable. The physical range and linguistic quantization are detailed below:

Variable Name	Role	Universe of Discourse (X)	Linguistic Terms	Unit
Engine Temperature	Input 1	[0, 120]	Cool, Normal, Hot	°C
Coolant Temperature	Input 2	[0, 120]	Low, Normal, High	°C
Environmental Temperature	Input 3	[0, 50]	Cold, Mild, Hot	°C
Radiator Fan Duration	Output	[0, 30]	Off, Short, Medium, Long	Minutes

3.3 Mathematical Formulation of Membership Functions

Piecewise linear triangular and trapezoidal membership functions are selected for computational efficiency on standard 16-bit and 32-bit automotive microcontrollers (e.g., Infineon TriCore / NXP S32K), avoiding floating-point intensive transcendental exponential functions required by Gaussian distributions.

Triangular Membership Function: $\mu(x; a, b, c) = \max(\min((x - a)/(b - a), (c - x)/(c - b)), 0)$

Trapezoidal Membership Function: $\mu(x; a, b, c, d) = \max(\min((x - a)/(b - a), 1, (d - x)/(d - c)), 0)$

Variable	Linguistic Term	Function Shape & Parameters (a, b, c, [d])	Physical Operating Range
Engine Temp (0–120°C)	Cool	Left-Trapezoidal / Triangular (a=0, b=0, c=70)	0°C to 70°C (Cold start / Warmup)
Engine Temp (0–120°C)	Normal	Triangular (a=50, b=80, c=100)	50°C to 100°C (Standard band)
Engine Temp (0–120°C)	Hot	Right-Trapezoidal / Triangular (a=85, b=115, c=120)	85°C to 120°C (Thermal stress band)
Coolant Temp (0–120°C)	Low	Left-Trapezoidal / Triangular (a=0, b=0, c=60)	0°C to 60°C (Below operating loop)
Coolant Temp (0–120°C)	Normal	Triangular (a=40, b=75, c=95)	40°C to 95°C (Optimum fluid circulation)
Coolant Temp (0–120°C)	High	Right-Trapezoidal / Triangular (a=80, b=110, c=120)	80°C to 120°C (Boil-over risk band)
Ambient Temp (0–50°C)	Cold	Left-Trapezoidal / Triangular (a=0, b=0, c=20)	0°C to 20°C (Sub-ambient / Winter)
Ambient Temp (0–50°C)	Mild	Triangular (a=10, b=25, c=38)	10°C to 38°C (Moderate climate)
Ambient Temp (0–50°C)	Hot	Right-Trapezoidal / Triangular (a=30, b=50, c=50)	30°C to 50°C (Scorching summer heat)
Fan Duration (0–30 min)	Off	Left-Triangular (a=0, b=0, c=5), Peak = 0 min	0 to 5 minutes (No active cooling)
Fan Duration (0–30 min)	Short	Triangular (a=3, b=8, c=15), Peak = 8 min	3 to 15 minutes (Light cooling assist)
Fan Duration (0–30 min)	Medium	Triangular (a=10, b=18,	10 to 25 minutes (Moderate

		c=25), Peak = 18 min	heat dump)
Fan Duration (0–30 min)	Long	Triangular (a=20, b=28, c=30), Peak = 28 min	20 to 30 minutes (Emergency/Heavy load)

3.4 Critical Overlap Zones & Safety Cushion Design

A foundational feature of this design is the deliberate calibration of overlap regions between adjacent fuzzy partitions. For instance, Coolant Temperature exhibits deliberate overlap between **80°C and 95°C** (where both 'Normal' and 'High' sets maintain non-zero membership). This overlap provides an active **thermal safety cushion**: it prevents the controller from shutting off the cooling fan prematurely while fluid thermal saturation remains elevated, ensuring smooth actuator damping without thermal overshoot.

4. Fuzzy Rule Base Formulation

The rule base represents the cognitive knowledge core of the controller, encoding expert automotive thermal management heuristics. Each multi-antecedent rule uses the Zadeh min-intersection operator to resolve the condition *IF (Engine is A) AND (Coolant is B) AND (Environmental is C) THEN (Fan Duration is D)*.

4.1 Core Rule Base Matrix

The table below details the eight representative rules formulated for thermal control, establishing explicit physical engineering rationales for each combination:

Rule ID	Engine Temp	Coolant Temp	Ambient Temp	Fan Duration	Engineering Rationale
R1	Hot	High	Hot	**Long**	Severe thermal saturation in extreme summer heat; maximum cooling duration mandatory.
R2	Hot	High	Mild	**Long**	Engine and fluid loop both dangerously hot; requires long continuous cooling despite mild air.

R3	Hot	Normal	Hot	**Medium**	High block temperature in high ambient heat, but fluid loop stable; moderate fan duration.
R4	Normal	High	Hot	**Medium**	Fluid loop saturated under hot ambient conditions; requires active heat rejection assist.
R5	Normal	Normal	Mild	**Short**	Equilibrium condition under mild ambient climate; light maintenance run to prevent creep.
R6	Cool	Low	Cold	**Off**	Cold start during freezing conditions; fan OFF to accelerate engine warmup to operating temp.
R7	Normal	Low	Cold	**Off**	Coolant fluid cold under freezing weather; zero fan assist needed to avoid oil over-cooling.
R8	Cool	High	Hot	**Short**	Transient heat soak or thermostat lag in hot weather; short burst restores fluid balance.

4.2 Mamdani vs. Sugeno (TSK) Architectural Justification

A critical design trade-off in intelligent control is selecting between Mamdani and Takagi-Sugeno-Kang (TSK) inference frameworks. For automotive radiator control, the Mamdani architecture is explicitly chosen based on the following comparative criteria:

System Dimension	Mamdani Inference (Selected)	Sugeno / TSK Inference
Consequent Representation	Fuzzy membership curve ('THEN Duration is LONG')	Mathematical function ('THEN Duration = a·T_eng + b')
Human Interpretability	Exceptional: Rules are intuitive, readable, and directly auditable by automotive calibration engineers.	Moderate to low: Linear polynomial equations obscure intuitive thermodynamic boundaries.
Output Surface Dynamics	Yields smooth, non-linear geometric control contours ideal for human-calibrated thermal damping.	Yields piecewise linear or polynomial surfaces requiring rigorous numerical coefficients.
Actuator Safety Compliance	Inherent bounding within predefined geometric universes; highly stable under sensor fault transitions.	Risk of polynomial divergence outside calibrated parameter operational envelopes.

5. Implementation & Worked Case Study

To validate the fuzzy cooling controller, an end-to-end numerical execution is performed on an empirical live telemetry snapshot captured during high thermal stress conditions:

> **TEST TELEMETRY SNAPSHOT: Engine Temp = 92°C | Coolant Temp = 88°C | Ambient Temp = 35°C**

5.1 Step 1: Real-Time Fuzzification

Crisp sensor measurements are substituted into the active membership function equations to determine the exact degree of truth (μ) for each overlapping linguistic set:

Sensor Parameter	Active Term	MF Definition (a, b, c)	Mathematical Substitution	Membership Degree (μ)
Engine Temp = 92°C	Normal	(50, 80, 100)	(100 - 92) / (100 - 80) = 8 / 20	**0.400**
Engine Temp = 92°C	Hot	(85, 115, 120)	(92 - 85) / (115 - 85) = 7 / 30	**0.233**
Coolant Temp = 88°C	Normal	(40, 75, 95)	(95 - 88) / (95 - 75) = 7 / 20	**0.350**

Coolant Temp = 88°C	High	(80, 110, 120)	(88 - 80) / (110 - 80) = 8 / 30	**0.267**
Ambient Temp = 35°C	Mild	(10, 25, 38)	(38 - 35) / (38 - 25) = 3 / 13	**0.231**
Ambient Temp = 35°C	Hot	(30, 50, 50)	(35 - 30) / (50 - 30) = 5 / 20	**0.250**

5.2 Step 2: Mamdani Rule Antecedent Evaluation & Implication

With fuzzified memberships established, the active rules in the knowledge base are matched against telemetry. Each rule's antecedent firing strength **w_i** is evaluated using the Zadeh intersection operator (**min**), which subsequently clips the rule's output membership function:

Rule ID	Antecedents Evaluated	Min-Operator Resolution	Firing Strength (w)	Clipped Output Term
R1	Hot AND High AND Hot	min(0.233, 0.267, 0.250)	**0.233**	Long (Clipped at 0.233)
R2	Hot AND High AND Mild	min(0.233, 0.267, 0.231)	**0.231**	Long (Clipped at 0.231)
R3	Hot AND Normal AND Hot	min(0.233, 0.350, 0.250)	**0.233**	Medium (Clipped at 0.233)
R4	Normal AND High AND Hot	min(0.400, 0.267, 0.250)	**0.250**	Medium (Clipped at 0.250)
R5	Normal AND Normal AND Mild	min(0.400, 0.350, 0.231)	**0.231**	Short (Clipped at 0.231)

5.3 Step 3: Aggregation of Clipped Output Sets

Because multiple rules point to the same output terms (e.g., both R1 and R2 command 'Long', while R3 and R4 command 'Medium'), the controller applies the Zadeh union operator (**max**) to consolidate clipping levels for each term:

- **Consolidated Output Term 'Short':** Governed solely by Rule R5 → Clipped at height w = 0.231.
- **Consolidated Output Term 'Medium':** Governed by max(w_R3, w_R4) = max(0.233, 0.250) → Clipped at height w = 0.250.
- **Consolidated Output Term 'Long':** Governed by max(w_R1, w_R2) = max(0.233, 0.231) → Clipped at height w = 0.233.

- **Term 'Off':** Not triggered by any active rule (w = 0.000).

Aggregated Envelope Function: μ_agg(y) = max(min(0.231, μ_Short(y)), min(0.250, μ_Medium(y)), min(0.233, μ_Long(y)))

6. Results & Defuzzification

The final phase transforms the aggregated continuous fuzzy envelope into an authoritative crisp control value representing the exact actuator run duration in minutes.

6.1 Centroid Defuzzification (Center of Gravity)

The mathematical standard for Mamdani controllers is continuous Centroid Defuzzification, which computes the center of area along the universe of discourse [0, 30] minutes:

$$y^* = \left(\int_{[0,30]} y \cdot \mu_{agg}(y)\, dy \right) / \left(\int_{[0,30]} \mu_{agg}(y)\, dy \right)$$

Evaluating the numerical integral across the combined trapezoidal geometries of Short, Medium, and Long clipped sets yields the rigorous result:

FINAL DEFUZZIFIED ACTUATOR COMMAND

Radiator Fan Duration y* = 16.72 Minutes

True Mamdani Centroid Defuzzification (Area-Weighted COG)

6.2 Critical Methodological Correction: Centroid vs. Weighted Peak Average

An essential technical contribution of this report is resolving a common engineering discrepancy observed in rapid heuristic calculations: the difference between a **true geometric centroid** and a simplified **weighted average of peak centers**.

Methodology	Mathematical Formula	Computed Value	Technical Assessment
Weighted Peak Average (Simplified Heuristic)	$y = (\sum w_i \cdot y_{peak}) / (\sum w_i)$ $= (0.231 \cdot 8 + 0.250 \cdot 18 + 0.233 \cdot 28) / (0.231+0.250+0.233)$	**19.79 min**	Inaccurate for Mamdani FIS: Ignores geometric base widths, overlap integration, and asymmetric support areas.
True Centroid Defuzzification (Validated Standard)	$y^* = (\int y \cdot \mu_{agg}(y)\, dy) / (\int \mu_{agg}(y)\, dy)$ Continuous integration across merged clipped geometry	**16.72 min**	Correct Reference Value: Accurately reflects area centers and overlap geometry between Short, Medium, and Long sets.

> **■ ENGINEERING RECONCILIATION & CORRECTION NOTE**
> The 19.79-minute figure stems from a simplified discrete weighted average of peak centers (8, 18, 28 min). Because the 'Short' set's broad base (3–15 min) and the 'Medium' set's overlap pull substantial area toward lower values, the true continuous area-weighted centroid settles precisely at 16.72 minutes. Implementing 19.79 minutes in firmware would run the fan ~18% longer than theoretically optimal, incurring unneeded parasitic electrical draw.

7. Engineering Analysis & Future Extensions

The performance characteristics and operational advantages of the designed fuzzy cooling controller are analyzed below, followed by actionable engineering roadmaps for next-generation deployment.

7.1 Systematic Thermal & Actuator Benefits

- **Elimination of Thermal Shock:** Rather than violently cycling between 0% and 100% duty cycles, the controller commands continuous durations that adapt dynamically to ambient temperature, protecting the engine block against steep thermal gradients and head gasket stress.
- **Parasitic Electrical Load Reduction:** By avoiding rigid full-duration runs when vehicle speed or ambient temperatures provide natural cooling, the system saves substantial alternator output, directly reducing engine brake-specific fuel consumption (BSFC).
- **Multi-Sensor Conflict Arbitration:** When engine telemetry signals conflict—such as a hot engine block (92°C) coupled with sub-zero ambient weather (5°C)—the fuzzy rule base prevents destructive overcooling, seamlessly arbitrating between opposing sensor inputs.

7.2 Project-Based Learning (PBL) Extension Opportunities

To extend the capabilities of the baseline 3-input architecture toward full automotive series-production readiness, four key engineering extensions are formulated:

- **Extension 1: Re-Fuzzification Under Extreme Thermal Saturation:** Test Case: Engine Temp = 105°C, Coolant Temp = 92°C, Ambient Temp = 20°C. In this regime, Engine Temp fires μ_Hot = 0.667 (climbing sharply toward overheat), Coolant Temp sits in the upper Normal/High overlap (μ_High = 0.400, μ_Normal = 0.150), while Ambient is strictly Mild (μ_Mild = 0.769). This telemetry activates heavy 'Long' duration cooling rules, demonstrating robust protection under high-load towing scenarios.
- **Extension 2: 4th Input Integration — Dynamic Ram-Air Vehicle Speed:** Integrate Vehicle Speed telemetry (km/h) with linguistic sets: Stopped [0, 15], City [10, 60], and Highway [50, 140]. At highway velocities (>80 km/h), dynamic ram-air flow through the grille delivers massive convective cooling. The rule base can be augmented with suppressive rules (e.g., 'IF Speed is Highway AND Coolant is Normal THEN Fan Duration is Off'), eliminating redundant fan operation when ram-air suffices.

- **Extension 3: Environmental Breakpoint Adaptation (Desert vs. Arctic Modes):** Implement adaptive software operating modes. In 'Desert Towing Mode', shift the 'Hot' engine threshold downwards (e.g., center at 80°C instead of 115°C) to induce early proactive heat rejection. In 'Arctic Mode', expand the 'Cold' boundary upwards to prevent radiator icing and cabin heater deficiency.
- **Extension 4: Genetic Algorithm (GA) & Neuro-Fuzzy (ANFIS) Tuning:** Replace manual trial-and-error breakpoint tuning with an automated Genetic Algorithm (GA) or Adaptive Neuro-Fuzzy Inference System (ANFIS). A GA optimization objective function $J = \int [w_1(T_{eng} - T_{target})^2 + w_2(P_{fan})^2] dt$ can autonomously adjust triangle vertex coordinates (a, b, c) across standardized driving cycles (WLTP/EPA), simultaneously minimizing engine thermal deviation and fan electrical power consumption.

7.3 Conclusion

The designed Mamdani Fuzzy Logic Controller successfully bridges human calibration expertise and embedded automotive control. By handling non-linear thermal dynamics and soft thresholds across three distinct telemetry streams, it eliminates the electromechanical cycling, thermal shock, and energy inefficiency of conventional on/off thermostats. The mathematical validation confirms a rigorous defuzzified fan duration of 16.72 minutes under heavy thermal load, providing an optimized, production-grade foundation for advanced automotive thermal management systems.